

Table 1. Initial Benchmark Results on the Unrefined Dataset

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv8n	0.734	0.692	0.728	0.375
YOLOv10n	0.635	0.688	0.671	0.339
YOLO11n	0.851	0.745	0.807	0.424
YOLO26n	0.836	0.738	0.793	0.461
YOLO26s	0.844	0.722	0.786	0.474

Table 2. Dataset Audit Summary

Metric	Before Cleaning	After Cleaning
Total Images	7759	7264
Total Labels	7759	7264
Missing Labels	0	0
Empty Labels	0	0
Corrupt Images	0	0
Invalid Bounding Boxes	175	0
Invalid Class IDs	0	0

Table 3. Single-Class Benchmark Results on the Clean Dataset (Test Set)

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLOv10n	0.665	0.698	0.688	0.349
YOLOv8n	0.745	0.772	0.738	0.375
YOLO11n	0.668	0.746	0.767	0.482
YOLO26n	0.859	0.797	0.866	0.556
YOLO26s	0.859	0.813	0.876	0.580

Table 4. YOLO26n Lightweight Refinement Results

Model	Epochs	Precision	Recall	mAP@0.5	mAP@0.5:0.95
YOLO26n	50	0.859	0.797	0.866	0.556
YOLO26n	100	0.919	0.844	0.915	0.608

Table 5. Two-Class (Plastic vs Non-Plastic) Results

Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Plastic (pla)	0.820	0.860	0.893	0.587
Non-plastic	0.783	0.654	0.736	0.457
Overall	0.801	0.757	0.815	0.522

Table 6. Five-Class Multiclass Results

Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Bottle	0.724	0.727	0.781	0.527
Food_pack	0.684	0.606	0.753	0.583
Bag	0.599	0.636	0.652	0.420
O_pla	0.780	0.804	0.831	0.539
Non_pla	0.776	0.720	0.774	0.471
Overall	0.724	0.727	0.781	0.527

Table 7. Seven-Class Multiclass Results

Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Bottle	0.788	0.819	0.881	0.645
O_bottle	0.663	0.435	0.518	0.375
Food_pack	0.708	0.667	0.758	0.609
Bag	0.683	0.851	0.860	0.636
O_bag	0.628	0.604	0.660	0.457

Class	Precision	Recall	mAP@0.5	mAP@0.5:0.95
O_pla	0.768	0.788	0.840	0.532
Non_pla	0.817	0.717	0.768	0.521
Overall	0.722	0.697	0.755	0.539

Table 8: Performance Trend with Increasing Number of Classes

Configuration	mAP@0.5
Single-Class	0.915
Two-Class	0.815
Five-Class	0.781
Seven-Class	0.755

Table 9: S1 vs S2 Comparison on the RiSID 5-Class Dataset

Setting	Precision	Recall	mAP@0.5	mAP@0.5:0.95
S1 (Scratch)	0.290	0.401	0.303	0.185
S2 (Pretrained COCO Weights)	0.724	0.727	0.781	0.527

LITERATURE REVIEW TABLE

Paper	Type	Model	Dataset	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Key Limitation
Chen and Zhu (2024)	Object Detection	Lightweight YOLOv5	ORCA Flow-Img Dataset	0.887	0.787	0.849	N/R	Detection performance decreased under strong lighting and water reflection conditions.
Fan et al. (2024)	Object Detection	SS-YOLOv8n	SLD Dataset	0.827	0.683	0.801	0.511	Complex aquatic backgrounds and semi-floating litter reduced robustness.
Di et al. (2025)	Object Detection	ES-YOLOv8	Self-made Floating Waste Dataset	0.920	0.876	0.933	0.750	Additional attention and fusion modules increased architectural complexity.
Hou and Yu (2025)	Object Detection	PBM-YOLO	FloW-Img + Self Dataset	0.900	0.858	0.907	0.481	Performance decreased under occlusion and adverse weather conditions.
Haldorai et al. (2024)	Object Detection	Modified SSD + ResNet	Acquafresh / Floating Garbage Dataset	N/R	N/R	0.94099	N/R	Higher computational cost reduced scalability on larger multiclass datasets.
Sumon et al. (2026)	Object Detection	RT-DETR + Faster R-CNN	FloW-Img	0.7647	0.8799	0.89081	0.4407	Glare, waves, foam, and illumination variation affected RGB-based detection.
Lee et al. (2025)	Detection + Tracking	YOLOv8n + DeepSORT	Stream Video Dataset	0.980	0.979	0.992	0.714	Tracking consistency decreased under dynamic water conditions.
Nunkhaw and Miyamoto (2024)	Detection + Tracking	YOLOv5 + DeepSORT	River-floating Waste Dataset	0.800	0.940	0.880	N/R	Small and submerged waste objects remained difficult to detect reliably.
Heller et al. (2025)	Detection + Tracking	YOLOv8m + Norfair	Bottle Detection Dataset	N/R	0.95–0.98	N/R	N/R	Performance decreased under glare and distant camera placement.
Xu et al. (2025)	Object Detection	YOLOv10-N	Riverine Litter Dataset	0.924	0.902	0.944	N/R	Dense vegetation regions affected detection consistency.
Tikász et al. (2025)	Object Detection	YOLOv8s	River Macroplastic Dataset	0.850	0.700	0.770	0.450	Occlusion and geographically limited datasets reduced reliability.
Zhao et al. (2024)	Object Detection	RBL-YOLO	Underwater Riverbed Dataset	0.860	0.700	0.800	N/R	Underwater turbidity and blur reduced detection reliability.
Kataoka et al. (2024)	Instance Segmentation	YOLOv8l Segmentation	River Surface Dataset	N/R	N/R	0.773	0.553	Water-level variation and false positives affected performance.
Jia et al. (2023)	Classification	DenseNet121 / SqueezeNet	TUD-GV Dataset	N/R	N/R	N/R	N/R	The framework focused only on image classification without localization support.
Huang et al. (2021)	Cleaning System	MF-USV	Real-time Lake Experiments	N/R	N/R	N/R	N/R	The framework relied on color filtering instead of deep learning detection.
PLADICS (Proposed)	Object Detection + Intelligent Collection	YOLO26n	Merged Clean RiSID + Riverine Plastic Dataset	0.919	0.844	0.915	0.608	Real-world rover deployment, embedded optimization, and long-duration outdoor validation are still ongoing.